

Summary:

I appreciate the effort taken by the authors to strengthen the manuscript and find the revised version much improved in terms of clarity and accessibility. In particular, the introduction and motivation are much easier to follow. I believe this manuscript is a useful contribution to the literature; however, I have a few remaining comments.

Major comments:

1. I appreciate the authors’ attempt to address previous comments on determining the appropriate bin width (p. 12). The authors note that bin widths need not be constant over time was quite useful. This new paragraph on bin width has been added under the section “Implications for research design” and I wonder if it would be more appropriately placed elsewhere. The text’s current location implies that bin width should be determined when designing the experiment, rather than during analysis. I worry that suggesting that bin width be determined during the design of the experiment could suggest (misleadingly) to some researchers that response times ought to be collected in discrete time, rather than continuous time; doing so would limit researchers’ ability to consider alternative bin widths later during analysis. I suggest renaming this section “Implications for research design and analysis” and making it clear that response time data still should be recorded in continuous time, if multiple bin widths are to be considered later during analysis.
2. Related to the above comment on bin width, I also think some discussion of the sensitivity of results and conclusions to the choice of bin width is both necessary and useful, given that this is a tutorial paper and choosing the width of the bin is a key step in modeling. The authors state generally that the “goal is to find the smallest bin width that is supported by the amount of data available”. This guidance may be difficult to translate to actionable steps, particularly by a researcher who is new to EHA. For example, what sort of metric is used to determine “support”? Is it noise in the estimated hazard function or another criterion? More specificity here could be useful, and I would recommend including a supplementary example that illustrates both choosing a bin width and any potential impact of that bin width on the conclusions.
3. Figure 2: In the plot of $ca(t)$, why is there a gap between the first and second orange triangles? That is, why is $ca(t)$ missing at 160ms?
4. P. 23 line 407-409: The text states, “First, because the first few bins often contain no responses, one has to select an analysis time window, i.e., a contiguous set of bins for which there is data for each participant.” What is meant here by “there is data for each participant”? And why is it important that the first few bins often contain no responses—isn’t a lack of response at early time points useful data? This sentence could benefit from being re-written for increased clarity.

5. P. 38 line 641-643: “We might pick a hazard ratio...[that] is much smaller than that previously observed”. Do you mean closer to 1, i.e., weaker than previously observed?

Minor comments:

1. P. 26 line 446: “bernoulli” should be “Bernoulli”
2. P. 27 line 486: I would add the section reference numbers (section 4.6.4) for Kurz 2023a (either to the main text or to the reference list) to help interested readers find the relevant information.
3. P. 27 line 489: If you first round a number to ‘digits = 1’ and then format using ‘nsmall = 2’, you may get a misleading value of 0 (i.e., 0.009 would appear as 0.00, when in fact 0.01 is more correct). This should be corrected.
4. P. 32 line 558: “R package lme4()” should be “R package lme4”, otherwise the reference appears to be for a function, rather than an R package. Same comment for “faux{” — “faux” would be clearer.